

# Simple Antenna Preamplifier for AM Radios



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AM radios are still used for capturing transmissions from long distances (DX reception). But AM radios normally require long (10-30m) outdoor antennae. That is easy if you live in a village but not always practical in urban environments. Antennae of 2-5m length are easy to install and maintain but the signal from them can be very weak.

If you have no option but to use a relatively short antenna, you can improve the AM reception by adding the proposed low-noise antenna preamplifier for AM signals. Its operation range is 150-1700 kHz, covering the long-wave (LW) and medium-wave (MW) ranges.

## Circuit and working

Fig. 1 shows the circuit of antenna preamplifier for AM radios. The input

collector current at least 2 mA. In most cases, 2-5mA current is appropriate and can be adjusted with resistor R3. Larger currents, e.g. 10 mA, may improve the volume but reduce the battery life. Capacitor C2 separates the preamplifier from the AM radio. The power supply is +5V from the regulator 7805 (IC1). AC or DC power supply in the range of 8-15V may be applied to connector CON1.

The working of the circuit is simple. The preamplifier circuit provides a gain higher than 10 in the LW and MW range. The amplified signals are fed to antenna input of the AM radio receiver for better reception. The circuit can also work for short-wave (SW) range, depending on the quality of practical implementation.

## Construction and testing

An actual-size, single-side PCB for the antenna preamplifier is shown in Fig. 2 and its component layout in Fig. 3. Assemble the circuit on a PCB to minimise assembly errors and save time. Carefully assemble the components and double-check for any overlooked error.

The antenna should be at least 2m

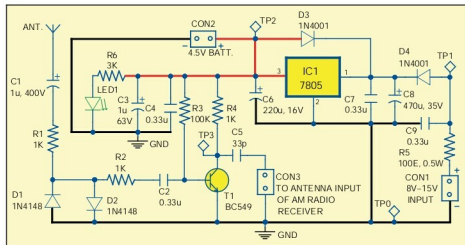


Fig. 1: Circuit of antenna preamplifier for AM radios

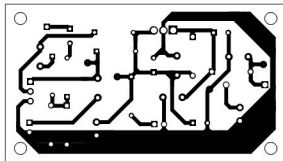


Fig. 2: Actual-size, single-side PCB for the antenna preamplifier

of the preamplifier is protected with capacitor C1, resistor R1, and diodes D1 and D2. Transistor T1 required for this application should be low-noise with moderate or high gain. For LW and MW ranges, BC550, BC547 and similar transistors will work well. It is important to keep the

## Test Points

| Test point | Details                        |
|------------|--------------------------------|
| TP0        | GND                            |
| TP1        | Input Supply                   |
| TP2        | 5V or 4.5V (when battery used) |
| TP3        | Amplified AM Signals           |

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